Near-Frontier with Reset (NFR) Planner: Formal Overview

ARC-AGI Abstraction Navigator Notes

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1 Problem Setting

Let $\mathcal{G} = (S, A, T)$ denote the agent interaction graph. States $s \in S$ encode masked ARC frames, actions $a \in A$ are constrained to the four arrows, and $T : S \times A \to S \cup \{\perp\}$ is the partial transition function estimated from experience ($\perp$ indicates an unknown successor).

Each play starts at a known level start state $s_0 \in S$. The planner is given:

- $S_{\text{disc}} \subseteq S$: discovered states from prior episodes.
- $T_{\text{known}} : S_{\text{disc}} \times A \to S_{\text{disc}} \cup \{\perp\}$: the partial transition map recovered from experience (returning $\perp$ when a transition has not yet been seen).
- $A(s) \subseteq A$: actions the environment currently allows at state s.
- $\mathcal{F} = \{s \in S_{\text{disc}} \mid \exists a \in A(s) : T_{\text{known}}(s, a) = \perp\}$, the *frontier* states whose outgoing edges are incompletely observed.

During an episode the agent sits in state s_t with available actions $A_t = A(s_t)$. The planner must output $a_t \in A_t \cup \{\text{RESET}\}$ to minimise the distance to a frontier while optionally using reset jumps to s_0.

2 High-Level Algorithm

We formalise the planner as follows:

$$D_t^c, P_t^c = \text{BFS}(\mathcal{G}_t, s_t), \tag{1}$$

$$D_t^0, P_t^0 = \text{BFS}(\mathcal{G}_t, s_0), \tag{2}$$

where $\mathcal{G}_t = (S_{\text{disc}}, A, T_{\text{known}})$ is the known subgraph. The superscript c records quantities measured from the *current* state $c \equiv s_t$, while 0 denotes measurements from the reset state s_0. The maps D store shortest-path lengths, and P store predecessors and the first action on the path.

For each frontier $f \in \mathcal{F}$, define the *reset-augmented cost*:

$$J(f) = \min\{D_t^c(f), 1 + D_t^0(f)\}. \tag{3}$$

Let $f^\star$ be the minimiser of J, breaking ties by smaller $D_t^{(0)}(f)$ and then lexicographically by state identifier.

The planner returns:

$$a_t = \begin{cases} \text{first\_action}(P_t^c, s_t, f^\star), & \text{if } D_t^c(f^\star) \leq 1 + D_t^0(f^\star), \\ \text{RESET}, & \text{otherwise.} \end{cases} \tag{4}$$

If the navigation policy yields an empty plan (i.e. $s_t = f^\star$ or the path is undefined), the agent probes the first $a \in A_t$ such that $T_{\text{known}}(s_t, a) = \perp$.

3 Desirable Properties

Completeness on the Known Subgraph. As long as $\mathcal{G}_t$ is connected between s_t and all frontiers, BFS ensures $D_t^c(f)$ equals the true shortest distance in $\mathcal{G}_t$. Therefore the navigation step reaches $f^\star$ in minimal steps without reset.

Reset Optimality. The cost (3) chooses between continuing from s_t or incurring one reset plus the path from s_0. This is optimal under the assumption that resetting teleports to s_0 in one step and costs a unit time penalty.

Frontier Prioritisation. Any frontier with strictly smaller $J(f)$ than all others will eventually be chosen. If exploration uncovers new edges, J automatically updates without retuning.

Idempotence of Known Actions. The planner never replays an action whose successor is already in $T_{\text{known}}(s_t, a)$ when probing frontiers; thus exploration focuses on unknown transitions.